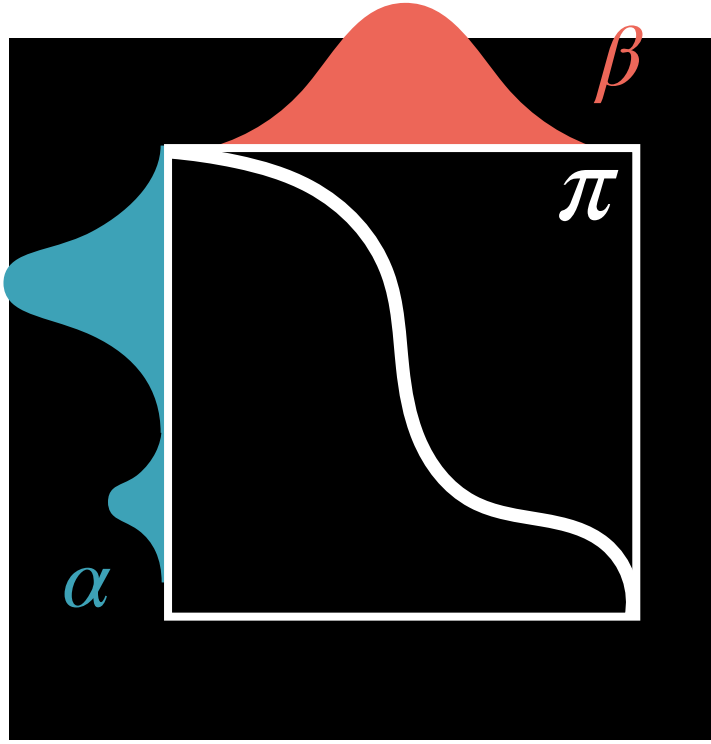
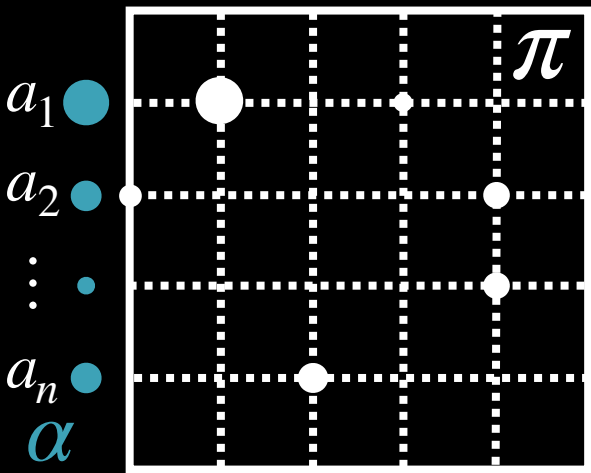
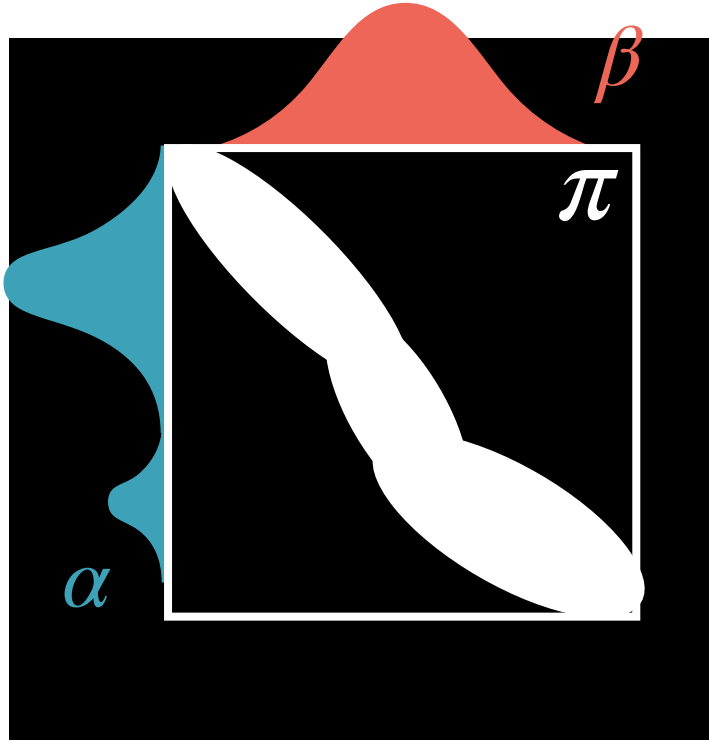




b_1 b_2 $\dots$ b_m β



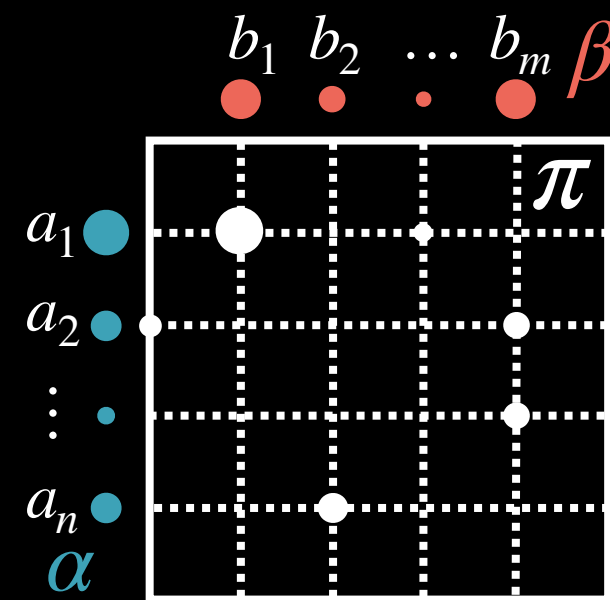
b



From Discrete to Continuous Couplings

Discrete

Represented by a matrix $\pi \in \mathbb{R}^{n \times m}$

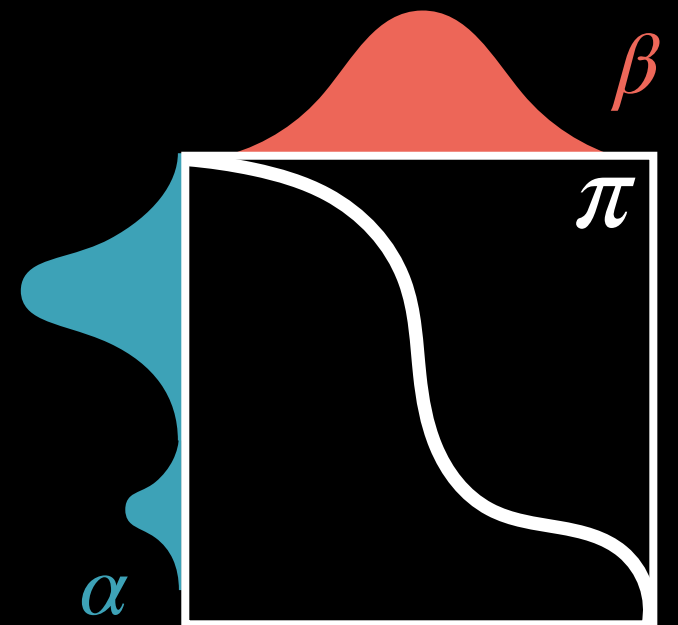


Each entry π_{ij} = amount of mass moved from point x_i to y_j

Constraints: row sums = $\mathbf{a}$, col sums = $\mathbf{b}$

Continuous

Represented by joint probability measure $\pi \in \mathcal{P}(X \times Y)$



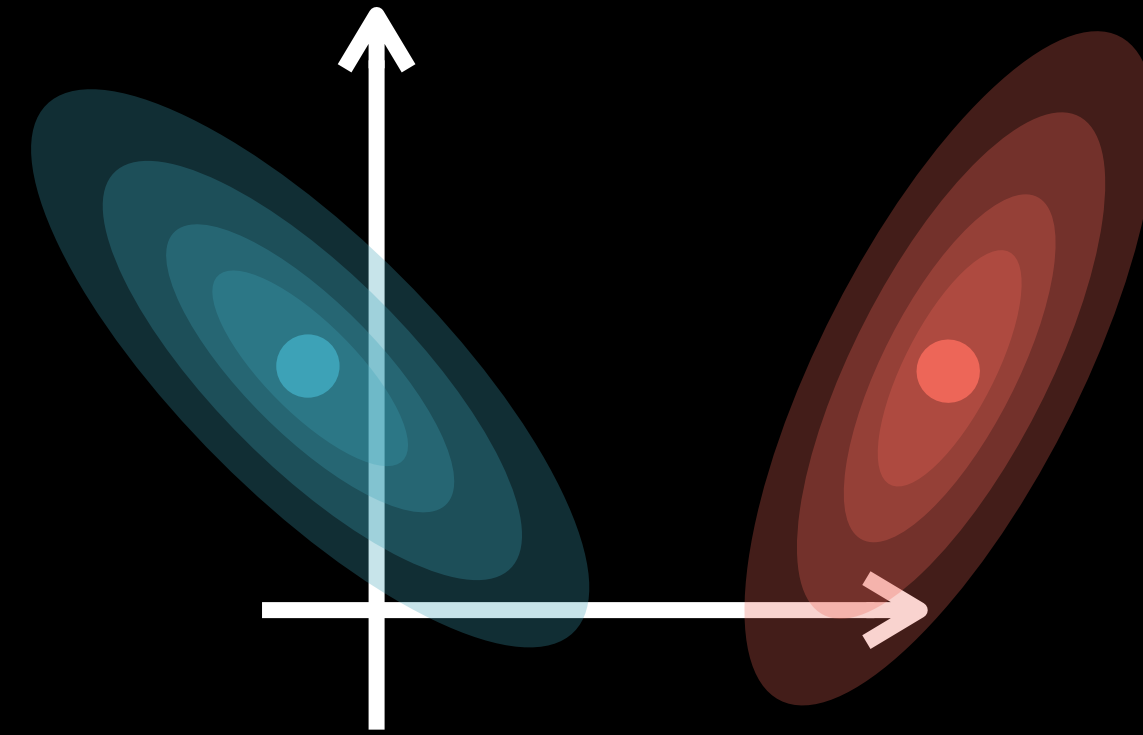
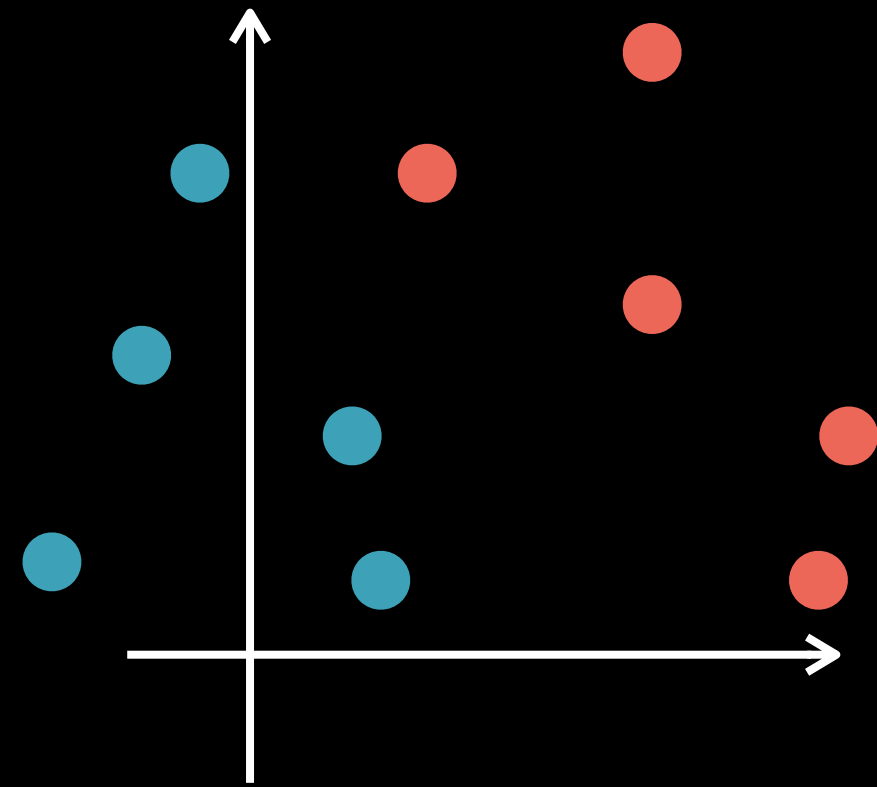
Each value $\pi(x, y)$ = infinitesimal density over pairs (x, y)

Constraints: $P_{\mathcal{X}\#}\pi = \alpha$, $P_{\mathcal{Y}\#}\pi = \beta$,
($P_{\mathcal{X}\#}$ is pushforward of projection $P_{\mathcal{X}}(x, y) = x$)

Equivalently:

$$\pi(A \times \mathcal{Y}) = \alpha(A), \pi(\mathcal{X} \times B) = \beta(B)$$

Kantorovich Formulation: General Case



Measures

$$\alpha = \sum_i a_i \delta_{x^{(i)}}, \beta = \sum_j b_j \delta_{y^{(j)}}; \quad \mathbf{a} \in \Delta_n, \mathbf{b} \in \Delta_m$$

Cost

$$\mathbf{C} \in \mathbb{R}^{n \times m}, \quad C_{ij} = c(x^{(i)}, y^{(j)})$$

Coupling

$$\pi = \begin{array}{c} \begin{array}{c} a_1 \\ a_2 \\ \vdots \\ a_n \\ \alpha \end{array} \begin{array}{|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet \\ \hline \bullet & \bullet & \bullet & \bullet \\ \hline \bullet & \bullet & \bullet & \bullet \\ \hline \bullet & \bullet & \bullet & \bullet \\ \hline \end{array} \end{array} \in \mathbb{R}^{n \times m}$$

Problem

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \Gamma, \mathbf{C} \rangle = \sum_{ij} \pi_{ij} c(x^{(i)}, y^{(j)})$$

Constraints

$$\Pi(\mathbf{a}, \mathbf{b}) = \{ \pi \mid \pi^\top \mathbf{1} = \mathbf{a}, \pi \mathbf{1} = \mathbf{b} \}$$